

1 Introduction

In this milestone we implemented from scratch a Multi-Layer Perceptron (MLP) and a K-Means classifier, and applied them to the Gaming and Mental Health dataset. The MLP is used for both classification (predicting the addiction level in {Low, Medium, High}) and regression (predicting the addiction score in $[0, 10]$), while K-Means is turned into a classifier by majority voting on its clusters. A central theme of this report is class imbalance: the dataset is strongly skewed, and we show how a weighted loss lets the MLP recover the rare class that a plain loss completely ignores.

2 Method

2.1 Data preparation

Features are standardised with the training-set mean and standard deviation, reused on the validation/test splits to avoid leakage; constant columns are guarded with $\sigma = 1$. We keep a 20% validation split for model selection and only use the test set for the final comparison. Our training split is heavily imbalanced: 896 Low, 340 Medium and only 44 High ($\approx 3\%$).

2.2 Multi-Layer Perceptron

We implemented the forward pass ($a_\ell = z_{\ell-1}W_\ell + b_\ell$, $z_\ell = f_\ell(a_\ell)$), back-propagation (chain rule, $\delta_{\ell-1} = (\delta_\ell W_\ell^\top) \odot f'_\ell(a_{\ell-1})$) and a mini-batch SGD update entirely by hand. Weights use the $\mathcal{N}(0, 1)/\sqrt{\text{fan_in}}$ initialisation. We coded the Sigmoid, ReLU and Identity activations and the MSE, Cross-Entropy and MAE (L1) losses. In all experiments we used a single-hidden-layer MLP, and selected the hidden dimension on the validation set. For classification the baseline uses a Sigmoid output trained with MSE on a one-hot target; for regression an Identity output with MSE.

2.3 Weighted cross-entropy for imbalance

Our baseline classifier uses a Sigmoid output trained with MSE on one-hot targets. To better handle class imbalance, we added a Softmax output with a *weighted* cross-entropy $\mathcal{L} = -\frac{1}{N} \sum_n \sum_k w_k t_{nk} \log(y_{nk})$, with per-class weights set to the inverse class frequency $w_k = N/(C N_k)$ (here $w = [0.48, 1.21, 10.4]$). The softmax+CE combination yields the clean output-layer gradient $w \odot (y_{\text{pred}} - y_{\text{true}})$.

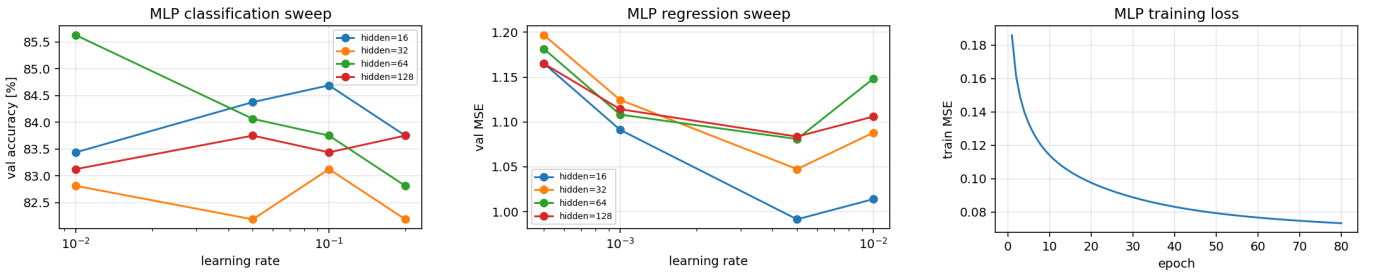
2.4 K-Means classifier

Standard K-Means: random initialisation, assign each point to the nearest centre, recompute centres as cluster means, repeat until the centres stop moving. Each cluster is then labelled by majority vote of its training points, and an unseen point is predicted with the label of its nearest centre. Empty clusters keep their previous centre.

3 Experiments and results

3.1 Hyperparameter selection

On the validation set we swept the MLP hidden size $\in \{16, 32, 64, 128\}$ and learning rate $\in \{0.01, 0.05, 0.1, 0.2\}$ (80 epochs, batch 32), retaining hidden=64, $\eta = 0.01$ (Fig. 1a). The training loss decreases smoothly and flattens, without overfitting (Fig. 1c). For regression we swept the same hidden sizes against smaller learning rates and kept hidden=16, $\eta = 0.005$, reaching validation MSE ≈ 0.99 (Fig. 1b). For K-Means we tested $K \in \{3, \dots, 100\}$ (averaged over 3 inits); both accuracy and macro-F1 keep improving with K up to $K = 100$ (Fig. 2c), which we retained.



(a) classification sweep

(b) regression sweep

(c) training loss

Figure 1: MLP hyperparameter sweeps and training-loss curve of the selected classifier.

3.2 Final test performance

Method	Classification (test)			Regression (test)	
	Acc.	Macro-F1	Fit [s]	Method	MSE
MLP baseline (sigmoid+MSE)	85.25%	0.546	0.14	MLP (h=16)	1.037
MLP softmax + weighted-CE	84.50%	0.757	0.14	MS1 Linear Reg.	0.994
K-Means ($K=100$)	78.25%	0.585	0.05		
MS1 Logistic Regression	85.75%	0.597	–		

Table 1: Final test performance, with our MS1 supervised baselines added for comparison.

The baseline MLP matches Logistic Regression on accuracy (85.25%) but, like every global-loss model, it collapses the rare class: its per-class recall is $[0.95, 0.70, 0.00]$ – *no High sample is ever predicted* (Fig. 2a). This is exactly why its macro-F1 stays low (0.55) despite a high accuracy: accuracy is dominated by the majority class, while macro-F1 averages all classes and is punished by the empty High row. The weighted cross-entropy fixes this: recall becomes $[0.92, 0.66, 0.69]$, the High class is now recovered (Fig. 2b), and macro-F1 jumps to **0.757** for only a 0.75% accuracy drop. This is the most useful model overall and clearly the best trade-off on this imbalanced task.

For regression the MLP (MSE 1.037) does not beat the MS1 closed-form linear model (0.994), which strongly suggests the addiction score is essentially a linear function of the features. K-Means with a large K gives a decent macro-F1 (0.585) by carving small clusters for the rare class, but its accuracy is the lowest.

Test: baseline (sigmoid+MSE)

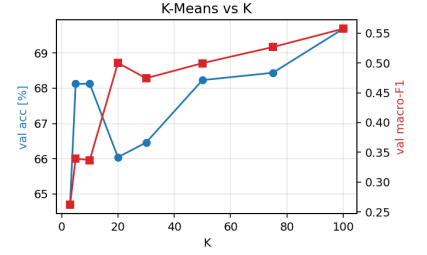
	Low	Medium	High
Low	266	14	0
Medium	32	75	0
High	0	13	0
	Low	Medium	High
	predicted		

(a) baseline (MSE)

Test: softmax + weighted CE

	Low	Medium	High
Low	258	22	0
Medium	31	71	5
High	0	4	9
	Low	Medium	High
	predicted		

(b) weighted-CE



(c) K-Means vs K

Figure 2: Test confusion matrices: the baseline never predicts High (a), the weighted loss recovers it (b). (c) K-Means accuracy/macro-F1 as a function of K .

3.3 Validation vs. test, and robustness

The baseline MLP reaches 85.6% validation and 85.25% test accuracy – almost identical, so the held-out estimate is reliable. To confirm this is not luck we ran a 5-fold cross-validation of the chosen architecture on the full training set, obtaining $84.2 \pm 0.7\%$ accuracy and 0.541 ± 0.006 macro-F1. The very small standard deviation shows the remaining val/test gap is just split variance (320 val / 400 test points), not a generalisation problem. We also complemented the main experiments with this cross-validation estimate, the MAE loss and the weighted cross-entropy variant, to better study robustness and class imbalance.

3.4 Runtime

The MLP trains in ≈ 0.14 s and predicts a 400-sample batch in < 0.1 ms. K-Means trains in ≈ 0.05 s but is slower at prediction (≈ 1.5 ms) because every point is compared to all K centres. Both are far cheaper at prediction time than the MS1 KNN, which compares against the whole training set.

4 Discussion and conclusion

Our main finding is that, on this imbalanced dataset, the choice of loss matters more than raw model capacity. A plain MLP matches logistic regression but silently ignores the minority class; switching to a weighted cross-entropy turns a 0.55 macro-F1 into 0.76 while keeping accuracy essentially unchanged, which is the behaviour we actually want when the rare class is the one of interest. For regression, a from-scratch MLP does not improve on a simple linear model, confirming a mostly-linear relationship. Overall, reporting accuracy together with macro-F1, per-class recall, a confusion matrix and a cross-validation estimate gives a far more honest picture than accuracy alone – and the weighted loss is a simple, principled way to act on what those metrics reveal.